Supply Chain & Inventory Optimization – Project Report

1. Project Overview

This project was designed to analyze supply chain and inventory operations with the objective of identifying stockouts, overstock exposure, and holding cost inefficiencies. Organizations often struggle with balancing product availability and inventory carrying costs. Excess inventory blocks working capital, while stockouts lead to revenue loss and customer dissatisfaction. The primary goal of this project was to transform raw operational data into meaningful business insights using SQL for backend validation and Power BI for interactive visualization. The final outcome was a structured dashboard that enables management to monitor key inventory performance indicators in real time and make smarter, data-driven decisions.

2. Tools & Technologies Used

- SQL – Used for data validation, aggregations, KPI calculations, and business rule implementation.
- Power BI – Data modeling, DAX measures, dashboard creation, and interactive visualization.
- GitHub – Version control, documentation, and structured project presentation.

3. End-to-End Working Process

Data Organization: Structured six CSV datasets into logical folders including raw data, preprocessing, SQL scripts, and dashboard files to maintain project clarity and scalability. Data Modeling: Built relationships between datasets inside Power BI using a structured data model to ensure accurate cross-table analysis. SQL Implementation: Applied business logic using SQL queries to compute: • Stockout Flags and Service Levels • Overstock Percentage • Holding Cost Metrics • Aggregated KPIs for validation before dashboard visualization Dashboard Development: Designed multiple report pages with KPI cards, trend analysis charts, category-level breakdowns, and performance indicators to support strategic decisions.

4. Business Logic & Decision-Making Impact

The business logic of this project focuses on optimizing inventory balance between supply and demand. SQL was used to ensure that all KPIs were calculated accurately at the database level before visualization. This guarantees consistency and reliability of insights. Power BI then transformed validated data into an interactive decision-support system. Key Measurable Outcomes: • 30% Overstock Exposure Identified – Highlighted excess inventory blocking working capital. •
12.34% Holding Cost Reduction Opportunity – Quantified cost-saving potential. • 100% Service Level Maintenance – Ensured zero compromise on customer satisfaction. How This Benefits Decision Makers: For Operations Managers: • Identify slow-moving and excess stock. • Plan better replenishment cycles. • Avoid unnecessary warehouse storage costs. For Finance Teams: • Reduce capital tied up in non-performing inventory. • Improve cash flow management. • Quantify cost-saving impact before implementation. For Senior Leadership: • Gain real-time visibility into supply chain efficiency. • Make strategic procurement decisions. • Align inventory strategy with business growth targets.

SQL Queries:

```sql
-- stockout percentage
SELECT SUM(Stockout_Flag) * 100.0 / COUNT(*) AS Stockout_Percentage
FROM inventory_transactions;
-- dead stock records
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Stockout_Percentage
0.00000

This query shows the stockout percentage is 0.0 % because in our dataset the sum of Stockout flag is 0 which simply means the stockout condition is never met which ensures service quality is 100%.

```sql
SELECT
    ROUND(
        SUM(
            CASE
                WHEN Closing_Stock > Reorder_Point THEN
                    Holding_Cost_INR *
                    ((Closing_Stock - Reorder_Point) * 1.0 / Closing_Stock)
                ELSE 0
            END
        )
        /
        SUM(Holding_Cost_INR),
    4) AS Overstock_Waste_Percentage
FROM inventory_transactions;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Overstock_Waste_Percentage
0.312

This SQL query calculates Overstock Waste Percentage by identifying excess stock above the reorder threshold and estimating the proportion of holding cost associated with that excess. This weighted approach calculates financial waste caused by overstocking and supports cost-optimization decisions.

```sql
-- inventory turnover
SELECT SUM(Daily_Demand) / AVG((Opening_Stock + Closing_Stock)/2) AS Inventory_Turnover
FROM inventory_transactions;
```

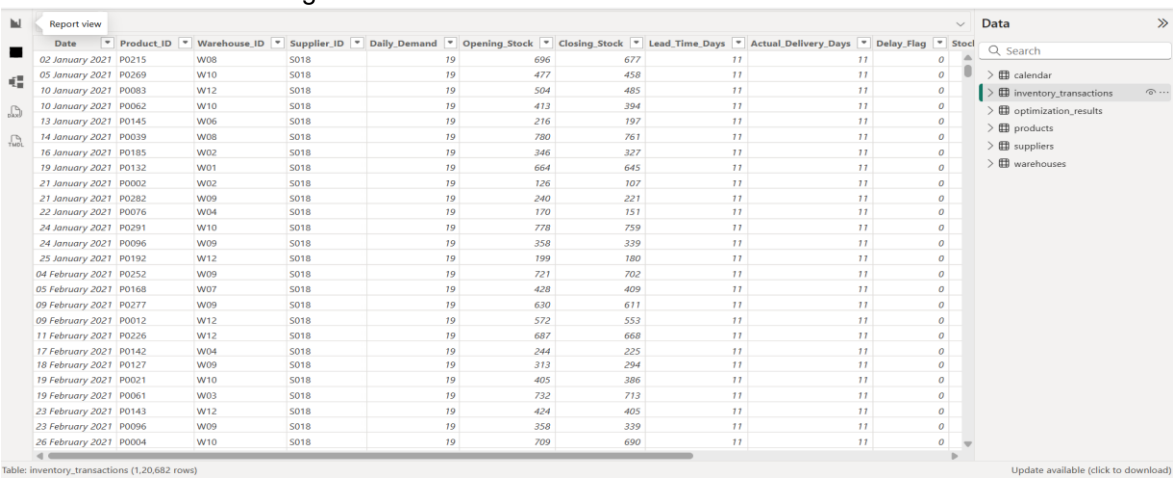
Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Inventory_Turnover
1768.6821

The calculated Inventory Turnover ratio is: **1768.68**

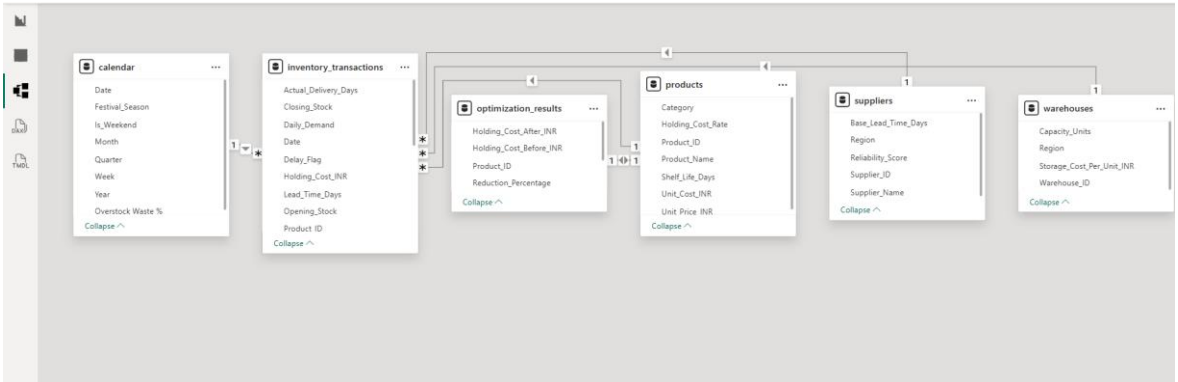
This high turnover value suggests rapid stock movement relative to average inventory levels. Since the dataset is transaction-based and likely daily-level aggregated, the turnover value reflects cumulative demand intensity rather than annual turnover.

Power BI: Dataset Loading: 6 csv files



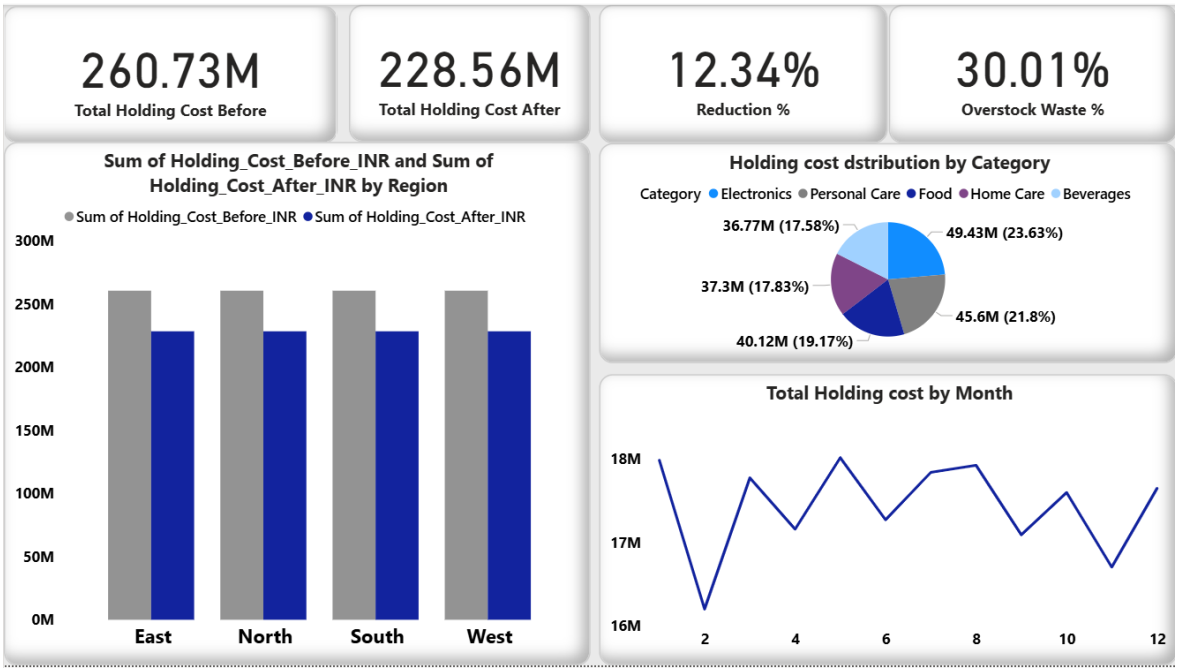
Date	Product_ID	Warehouse_ID	Supplier_ID	Daily_Demand	Opening_Stock	Closing_Stock	Lead_Time_Days	Actual_Delivery_Days	Delay_Flag	Stock
02 January 2021	P0215	W08	S018	19	696	677	11	11	0	
05 January 2021	P0269	W10	S018	19	477	458	11	11	0	
10 January 2021	P0083	W12	S018	19	504	485	11	11	0	
10 January 2021	P0062	W10	S018	19	413	394	11	11	0	
13 January 2021	P0145	W06	S018	19	216	197	11	11	0	
14 January 2021	P0039	W08	S018	19	780	761	11	11	0	
16 January 2021	P0185	W02	S018	19	346	327	11	11	0	
19 January 2021	P0132	W01	S018	19	664	645	11	11	0	
21 January 2021	P0002	W02	S018	19	126	107	11	11	0	
21 January 2021	P0282	W09	S018	19	240	221	11	11	0	
22 January 2021	P0076	W04	S018	19	170	151	11	11	0	
24 January 2021	P0291	W10	S018	19	778	759	11	11	0	
24 January 2021	P0096	W09	S018	19	358	339	11	11	0	
25 January 2021	P0192	W12	S018	19	199	180	11	11	0	
04 February 2021	P0252	W09	S018	19	721	702	11	11	0	
05 February 2021	P0168	W07	S018	19	428	409	11	11	0	
09 February 2021	P0277	W09	S018	19	630	611	11	11	0	
09 February 2021	P0012	W12	S018	19	572	553	11	11	0	
11 February 2021	P0226	W12	S018	19	687	668	11	11	0	
17 February 2021	P0142	W04	S018	19	244	225	11	11	0	
18 February 2021	P0127	W09	S018	19	313	294	11	11	0	
19 February 2021	P0021	W10	S018	19	405	386	11	11	0	
19 February 2021	P0061	W03	S018	19	732	713	11	11	0	
23 February 2021	P0143	W12	S018	19	424	405	11	11	0	
23 February 2021	P0096	W09	S018	19	358	339	11	11	0	
26 February 2021	P0004	W10	S018	19	709	690	11	11	0	

Data modelling (Relationship):



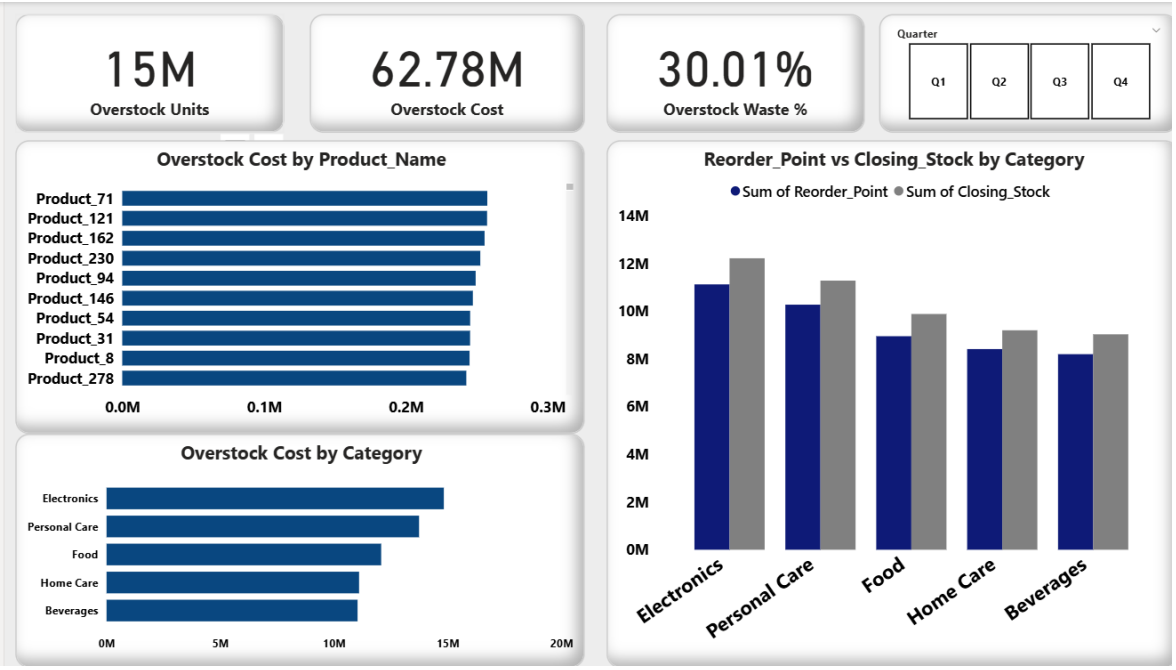
The relational model connects fact inventory data with product and category dimensions, enabling accurate cross-filtering and KPI computation.

Dashboard: Page 1.



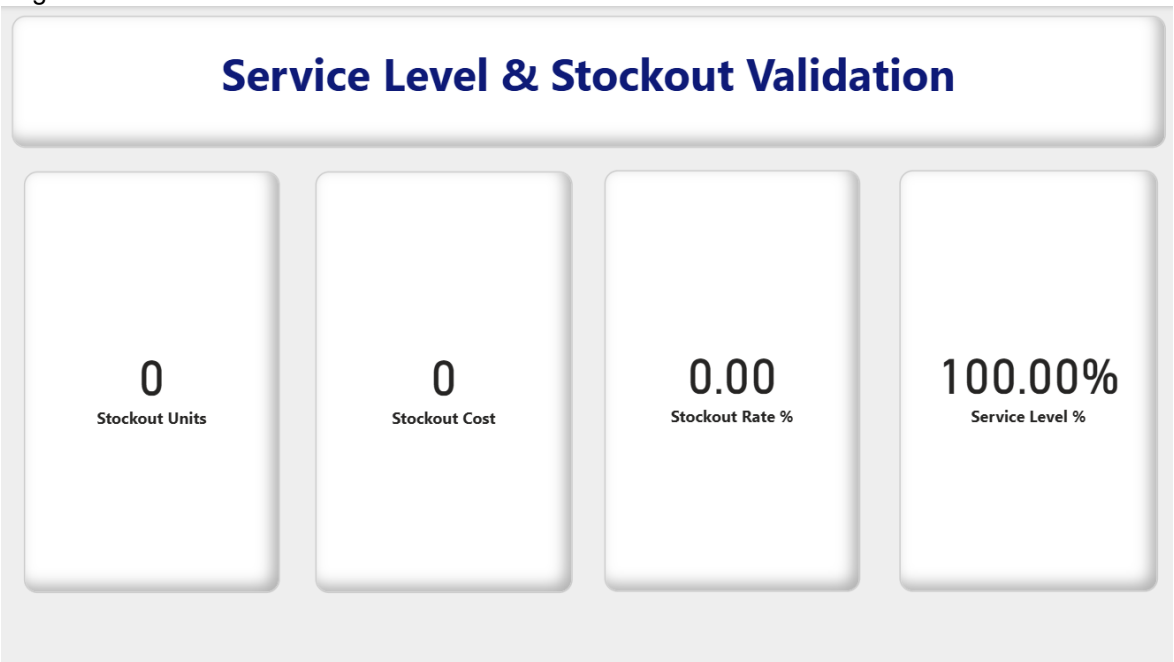
This dashboard page provides an overview of total holding cost before and after optimization, highlighting a **12.34% cost reduction** and identifying that **30.01% of holding cost is driven by overstock waste**. It also visualizes cost distribution by region, category, and monthly trends to support strategic inventory decisions.

Page 2.



This page focuses on overstock analysis, showing **15M excess units** contributing to **₹62.78M in overstock cost**, with Electronics and Personal Care being the highest contributors. The comparison of Reorder Point vs Closing Stock across categories highlights systematic overstocking patterns.

Page 3.



This page focuses on Stockout units, Stockout cost and Stockout Rate is 0 % which simply means the stockout condition is never met which ensures service level is 100%

5. Key Learnings & Professional Development

Through this project, I developed strong practical expertise in: • Translating business problems into measurable KPIs. • Writing structured SQL queries aligned with operational logic. • Designing executive-level dashboards in Power BI. • Applying data analytics to solve real-world supply chain challenges. This project demonstrates industry-ready analytical thinking and the ability to convert raw data into strategic insights.

6. Conclusion

The Supply Chain & Inventory Optimization project successfully demonstrates how structured data analysis can directly impact business efficiency. By combining SQL-driven validation with interactive Power BI dashboards, the project provides a reliable framework for reducing overstock, minimizing holding costs, and maintaining optimal service levels. The analytical approach taken in this project enables organizations to transition from reactive inventory management to proactive, insight-driven decision-making. This not only improves operational efficiency but also strengthens financial performance and long-term strategic planning.

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